

Shubh Kapoor

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TECHNICAL SKILLS

AI/ML and Deep Learning: Python, PyTorch, TensorFlow, scikit-learn, CNNs, multi-agent orchestration

Computer Vision: OpenCV, YOLOv8, MediaPipe, dlib, real-time inference pipelines

LLMs and RAG: RAG, HybridRAG, GraphRAG, Agentic RAG, ReAct agents, function calling, streaming responses, LangChain, LangGraph, LlamaIndex, HuggingFace, Groq, Ollama

Vector and Graph DBs: ChromaDB, Neo4j, FAISS, Pinecone, Weaviate, vector search, hybrid search, RAGAS, pytest

MLOps and Deployment: FastAPI, REST APIs, Docker, AWS Bedrock, MLflow, CI/CD, Git, GitHub, SQLite

Frontend: React.js, JavaScript ES6+, Tailwind CSS, HTML5, CSS3, Three.js, GSAP

Design: Figma, Adobe XD, Photoshop, Illustrator, After Effects

PROJECTS

Sentinel-47: AI Energy Supply Risk Intelligence Platform *Next.js, React, TypeScript, Gemini 2.5 Flash, Zustand, TanStack Query, SQLite, Tailwind CSS* [Live Demo](#)

- Engineered a multi-agent geopolitical intelligence platform integrating live market telemetry, corridor risk assessment, disruption simulation, and procurement optimization through four coordinated AI modules.
- Developed an LLM-powered scenario modelling engine with historical backtesting, weighted composite risk analysis, streaming executive memo generation, cryptographic audit ledger, and resilient offline fallback architecture.
- Implemented enterprise-grade frontend using Zustand state management, TanStack Query caching, deep-linkable dashboard sessions, guided demo workflows, error boundaries, and responsive visualization pipelines.

Healthcare Cost Intelligence *Python, SQLite, ChromaDB, sentence-transformers, FastAPI, Vercel* [Live Demo](#)

- Built an offline-first AI clinical decision tool taking patient inputs (symptom, city, age, comorbidities, budget) and returning explainable 5-component cost estimates with confidence scores, zero paid API dependencies.
- Engineered a rule-based intent mapper across 50+ procedures with city-tier, age, comorbidity, and ICU multipliers; applied semantic retrieval via ChromaDB and HuggingFace sentence-transformers across 43 hospital records.
- Designed a weighted multi-factor ranking algorithm covering clinical match, reputation, accessibility, and affordability; exposed results via a FastAPI REST endpoint and deployed the full stack to Vercel.

WORK EXPERIENCE

AI/ML Intern

May 2026 – Aug 2026

FintronicsAI Pvt. Ltd.

- Built a real-time driver drowsiness detection system using YOLOv8 and MediaPipe Face Mesh; computed Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR) thresholds to classify alert, drowsy, and sleeping states with high confidence.
- Developed an OpenCV inference pipeline for continuous webcam frame processing with alarm triggering on drowsiness detection and timestamped fatigue event logging for post-session data analysis.
- Benchmarked lightweight model variants (YOLOv8n vs MediaPipe) to optimise the accuracy-latency tradeoff for CPU-only deployment, applying model evaluation metrics to guide the final architecture decision.

UI/UX Designer Intern

Dec 2025 – Jan 2026

TalentiQo Workforce Pvt. Ltd.

- Delivered wireframes and high-fidelity Figma screens for 2 live product features across 3 review cycles; built a 20-component auto-layout library that standardised spacing, type, and interaction states across the design system.
- Authored developer-ready handoff specs covering breakpoints, hover/focus states, and edge-case flows, directly reducing engineering back-and-forth during implementation.

EDUCATION

The LNM Institute of Information Technology

Jaipur, India

B.Tech in Communication and Computer Engineering

2023 – 2027

Dr. Virendra Swarup Education Centre

Kanpur, India

High School, Science (ISC)

2021 – 2023